Semi-Supervised DeepSDF with Eikonal Regularization and Fourier Positional Encoding

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***Abstract*—Neural implicit representations such as DeepSDF learn continuous signed distance functions for 3D shape reconstruction, but require dense signed distance supervision for every training point. We investigate whether Eikonal regularization can compensate for reduced supervision, and whether Fourier positional encoding (PE) helps or hurts in this regime. Using a corrected preprocessing pipeline, we train on 299 ShapeNet shapes (airplane, chair, table) at supervision ratios from 5% to 100% and evaluate with Chamfer Distance (CD) and Normal Consistency (NC) on training-set reconstructions. The main result is that Eikonal regularization improves CD-based label efficiency, with the strongest reduced-label operating point at 50% supervision (EXP-03: CD = 0.0288, NC = 0.748), while a 3-seed 10% run remains numerically competitive but visually less reliable. In contrast, Fourier PE primarily collapses normal consistency at the 10% supervision slice we were able to retrain, while leaving CD broadly comparable to the no-PE baseline; lowering the encoding frequency from $L = 6$ to $L = 4$ partially recovers NC at a clear CD cost. We therefore conclude that Eikonal regularization is the useful ingredient for reducing signed-distance labels in this DeepSDF setup, whereas Fourier PE is not recommended at 10% supervision.**

***Index Terms*—DeepSDF, signed distance functions, Eikonal regularization, semi-supervised learning, positional encoding, 3D reconstruction**

I. Introduction

Learning 3D shape representations from data is a central problem in computer vision with applications in robotics, autonomous driving, and virtual reality. Traditional approaches such as voxel grids and point clouds suffer from resolution limitations and lack of surface continuity. Neural implicit representations address these issues by parameterizing shapes as continuous functions—typically signed distance functions (SDFs)—that can be queried at arbitrary resolution [1].

DeepSDF [1] learns a neural network $f_\theta(z, x) \to s$ that maps a latent code z and a 3D coordinate x to a signed distance value s. Through an auto-decoder framework, per-shape latent codes are jointly optimized with the network weights, enabling the model to represent a distribution of shapes. However, DeepSDF requires dense signed distance supervision: every training point must have a ground-truth SDF value, which is expensive to obtain for real-world 3D data.

This supervision bottleneck motivates semi-supervised approaches that leverage geometric priors to reduce label requirements. The Eikonal equation $\|\nabla_x f(x)\| = 1$ is a fundamental property of signed distance functions [2]. Enforcing this constraint on unsupervised points—those without SDF labels—provides a training signal derived purely from the geometry of the problem. If effective, this could substantially reduce the annotation effort for training neural implicit representations.

Separately, Fourier positional encoding (PE) has been shown to help multi-layer perceptrons (MLPs) learn high-frequency functions by mapping low-dimensional inputs to higher-dimensional sinusoidal features [3], [4]. While this technique is standard in Neural Radiance Fields (NeRF), its interaction with SDF learning—particularly in semi-supervised settings—is less understood.

In this work, we systematically investigate two questions:

1) Can Eikonal regularization reduce the need for signed distance supervision in DeepSDF, and by how much?
2) Does Fourier positional encoding help or hurt reconstruction quality in this semi-supervised SDF setting?

We report only the corrected post-fix experiments on 299 ShapeNet shapes (airplane, chair, table); earlier runs used a thin near-surface-only sampling band and are excluded throughout. Our key findings are:

- **Eikonal regularization improves CD-based label efficiency**: relative to the non-Eikonal baseline (EXP-01, CD = 0.0361, NC = 0.729), Eikonal improves full-label CD (EXP-02, CD = 0.0295) while slightly reducing NC, and keeps 50%, 10% and 5% supervision numerically competitive.
- **The 10% no-PE regime is mixed rather than definitive**: the 3-seed run is numerically competitive but qualitative inspection is less convincing than the metrics alone suggest, so we do not treat 10% supervision as the strongest visual operating point.
- **Fourier PE degrades NC at 10% supervision** while leaving CD close to the no-PE baseline; lowering L from 6 to 4 partially recovers NC at a clear CD cost.

II. Related Work

A. Neural Implicit Representations

Neural implicit representations encode 3D geometry as the level set of a learned function. Occupancy Networks [8] predict binary inside/outside occupancy, while DeepSDF [1]

regresses continuous signed distance values, enabling gradient-based surface extraction via marching cubes [7]. DeepSDF's auto-decoder framework avoids the need for an encoder by directly optimizing per-shape latent codes during training. While effective, it requires dense SDF supervision at every training point.

B. Geometric Regularization

The Eikonal equation $\|\nabla_x f(x)\| = 1$ is a necessary condition for valid signed distance functions. Gropp et al. [2] introduced Implicit Geometric Regularization (IGR), demonstrating that enforcing this constraint during training improves surface quality and enables learning from raw point clouds without SDF labels. This approach has been adopted in subsequent work on neural implicit surfaces [9], [10]. In our work, we investigate whether Eikonal regularization can reduce the required number of supervised training points, assessing its effects across multiple supervision ratios.

SIREN [5] takes a different approach to geometric regularization by using periodic activation functions (sinusoidal) instead of ReLU, which naturally satisfies certain differential equation constraints. However, SIREN requires architectural changes, whereas Eikonal regularization can be applied as a loss term to standard ReLU networks.

C. Positional Encoding

Fourier positional encoding was introduced in NeRF [3] to overcome the spectral bias of MLPs—their tendency to learn low-frequency functions first [11]. By mapping input coordinates through sinusoidal functions at multiple frequency bands, $\gamma(x) = [\sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)]$, the network can represent high-frequency variations. Tancik et al. [4] provided a theoretical framework for this approach based on the Neural Tangent Kernel.

While Fourier features are standard in radiance field methods, their application to SDF learning is less established. NeRF operates in a supervised regime where every ray has a target color, and the network is evaluated primarily at points along training rays. In contrast, SDF evaluation via marching cubes requires querying the network on a dense grid that includes regions far from the training data distribution, creating an extrapolation challenge for high-frequency encodings.

III. Method

A. Problem Formulation

Given a dataset of 3D shapes, we aim to learn a function $f_\theta : \mathbb{R}^{d_z} \times \mathbb{R}^3 \to \mathbb{R}$ that maps a shape-specific latent code $z_i \in \mathbb{R}^{d_z}$ and a spatial coordinate $x \in \mathbb{R}^3$ to the signed distance to the shape's surface. For each shape i, we have:

- **Supervised points**: $\{(x_j, s_j)\}_{j=1}^{N_s}$ with ground-truth SDF values (see Section IV-A for the exact sampling procedure)
- **Unsupervised points**: $\{x_k\}_{k=1}^{N_u}$ with coordinates only

During preprocessing, each shape is sampled to produce a fixed pool of 250,000 supervised points and a separate pool of 250,000 unsupervised points. The supervision ratio r controls the fraction of the supervised pool that is retained: at $r = 0.1$ only 25,000 supervised points are used, while all 250,000 unsupervised points remain available. At $r = 1.0$ the full supervised pool is used alongside the full unsupervised pool. Each training batch samples half its points from the supervised set and half from the unsupervised set. We investigate $r \in \{0.05, 0.10, 0.50, 1.0\}$.

B. Architecture

We use an 8-layer MLP with hidden dimension 512, following the DeepSDF architecture [1]. A skip connection concatenates the input to the activations at layer 4. Each shape is associated with a learnable latent code $z_i \in \mathbb{R}^{256}$, initialized from $\mathcal{N}(0, 0.01^2)$ and jointly optimized with the network weights (auto-decoder). The final layer produces a scalar SDF value without activation, with weights initialized from $\mathcal{N}(0, 1/\sqrt{512})$ and bias set to 0.1 to approximate a sphere.

When Fourier PE is enabled, input coordinates are transformed as:

$$\gamma(x) = [\sin(2^0\pi x), \cos(2^0\pi x), \ldots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x)] \tag{1}$$

where L is the number of frequency levels. This maps each 3D coordinate from $\mathbb{R}^3$ to $\mathbb{R}^{6L}$. The network input becomes $[z_i; \gamma(x)]$ instead of $[z_i; x]$. We test $L = 4$ (24-dimensional) and $L = 6$ (36-dimensional).

C. Loss Functions

The total training loss combines four terms:

$$\mathcal{L} = \mathcal{L}_{\text{sdf}} + \lambda_{\text{eik}}(t) \cdot \mathcal{L}_{\text{eik}} + \lambda_z \cdot \mathcal{L}_z + \lambda_{\text{2nd}} \cdot \mathcal{L}_{\text{2nd}} \tag{2}$$

SDF loss ($\mathcal{L}_{\text{sdf}}$): L1 regression on supervised minibatch points only:

$$\mathcal{L}_{\text{sdf}} = \frac{1}{N_s} \sum_{j=1}^{N_s} |f_\theta(z_i, x_j) - s_j| \tag{3}$$

Eikonal loss ($\mathcal{L}_{\text{eik}}$): Enforces $\|\nabla_x f\| = 1$ on all points (supervised and unsupervised), providing geometric regularization from unlabeled data:

$$\mathcal{L}_{\text{eik}} = \frac{1}{N_s + N_u} \sum_k (\|\nabla_x f_\theta(z_i, x_k)\| - 1)^2 \tag{4}$$

Latent regularization ($\mathcal{L}_z$): Prevents latent code drift:

$$\mathcal{L}_z = \|z_i\|^2 \tag{5}$$

Second-order loss ($\mathcal{L}_{\text{2nd}}$): Optionally penalizes the divergence of the gradient field on the same minibatch points as $\mathcal{L}_{\text{eik}}$ to encourage smoother surfaces:

$$\mathcal{L}_{\text{2nd}} = \frac{1}{N_s + N_u} \sum_k |\nabla \cdot \nabla_x f_\theta(z_i, x_k)| \tag{6}$$

This term is implemented in the codebase but not exercised in the corrected post-fix experiment matrix; the planned EXP-08 ablation was not retrained.

D. Eikonal Warmup Schedule

The Eikonal weight uses a linear warmup to prevent the gradient norm constraint from dominating before the network learns basic geometry from $\mathcal{L}_{\text{sdf}}$:

$$\lambda_{\text{eik}}(t) = \lambda_{\text{eik}} \cdot \min\left(1, \frac{t}{T_w}\right) \tag{7}$$

where T_w is ratio-dependent: 100 epochs for $r \geq 0.5$, 150 epochs for $r = 0.1$, and 200 epochs for $r = 0.05$. Lower supervision ratios require longer warmup because fewer supervised points provide weaker initial geometry signal.

E. Training Procedure

Training iterates over all shapes per epoch, sampling a batch of 16,384 points per shape split evenly between supervised and unsupervised samples. We use the Adam optimizer with learning rate 5×10^{-4}, decayed by 0.5× every 500 epochs, and gradient clipping at `max_norm` = 1.0.

Default hyperparameters are $\lambda_{\text{eik}} = 0.1$, $\lambda_z = 10^{-4}$, and $\lambda_{\text{2nd}} = 0$ (disabled). EXP-01 and EXP-02 train for 3,000 epochs; the post-fix priority retrains (EXP-03/04/05/06/11, including the EXP-04 and EXP-06 multi-seed replicates) train for 1,500 epochs to fit the QoS budget.

IV. Experiments

A. Dataset

We use 300 shapes from ShapeNet [6] spanning three categories (100 airplanes, 100 chairs, 100 tables). Preprocessing normalizes each mesh to the unit sphere and then samples two point pools per shape:

- **Supervised pool** (250,000 points). An 80/20 mix:
  - *Near-surface (200K)* points generated by multi-scale normal offsets at scales $\epsilon \in \{0.005, 0.01, 0.05\}$ with multipliers $\{-2, -1, 1, 2\}$, yielding SDF magnitudes up to $|s| = 0.1$.
  - *Far-field (50K)* points sampled uniformly inside the unit sphere with exact signed distance from `trimesh.proximity.signed_distance`. Trimesh's convention is positive-inside/negative-outside; we invert the sign to match the project convention (positive-outside/negative-inside).

  Meshes with more than 15,000 faces are decimated to 10,000 faces before the far-field query to keep preprocessing tractable.
- **Unsupervised pool** (250,000 points). Uniform in the unit sphere, coordinates only; used by the Eikonal term.

Each mesh is processed with a deterministic per-mesh seed derived from the global seed and the mesh name, so parallel preprocessing is reproducible. One airplane mesh (airplane_0077) had degenerate triangles that caused `signed_distance` to stall beyond 30 minutes; it was dropped from the pipeline, leaving 299 shapes. All 299 shapes are used for training (`train_split` = 1.0); we do not hold out a separate test set in the post-fix pipeline, because the experiment target is a stress test of label efficiency and PE interactions on the training objective rather than generalization to unseen shapes. Reported CD/NC are therefore reconstruction metrics over the training set, and should be interpreted accordingly.

For reduced-supervision runs, supervised points are subsampled to the target ratio (e.g. 10% retains 25,000 labeled points) while all 250,000 unsupervised points remain available for Eikonal regularization.

TABLE I

Final corrected post-fix experiment set actually used in this paper. All experiments use Eikonal regularization except EXP-01. Originally-planned slots EXP-07/08/09/10/12 were not retrained under the post-fix QoS budget and are excluded from all tables, figures, and conclusions.

ID	r	Eik	PE	Seeds	Purpose
01	100%	–	–	42	Baseline
02	100%	✓	–	42	Full-label Eikonal
03	50%	✓	–	42	Half labels
04	10%	✓	–	42,123,456	Low-label no-PE
05	5%	✓	–	42	Minimal labels no-PE
06	10%	✓	L=6	42,123,456	Low-label PE
11	10%	✓	L=4	42	Lower PE freq, low labels

B. Evaluation Metrics

For each trained model we extract a mesh per shape via marching cubes [7] on the SDF field clipped to the unit sphere. All post-fix evaluations are run at resolution 128^3. We report two metrics over the 299 training shapes:

Chamfer Distance (CD): L1 bidirectional mean nearest-neighbor distance between 30,000 points sampled from predicted and ground-truth meshes. Lower is better.

Normal Consistency (NC): Mean absolute dot product between normals at matched nearest-neighbor pairs. Higher is better (maximum 1.0).

Volumetric IoU and Eikonal Deviation were part of the original plan but were not computed; we discuss this explicitly in Section VI-C.

C. Experiment Design

Table I summarizes the final corrected post-fix experiment matrix used in this report. We vary supervision ratio ($r \in \{5\%, 10\%, 50\%, 100\%\}$), Eikonal regularization, and Fourier PE frequency ($L \in \{4, 6\}$). Key comparisons are EXP-03 vs. EXP-01/02 (label reduction without PE), EXP-04 vs. EXP-03 (what is lost when supervision drops from 50% to 10%), EXP-06 vs. EXP-04 (PE effect at 10% supervision), and EXP-11 vs. EXP-06 (PE frequency follow-up at 10% supervision). Two configurations were multi-seeded (seeds 42, 123, 456): EXP-04 and EXP-06. Their coefficients of variation remain below the project threshold of 0.2, so no seed expansion was required.

All experiments were trained on a single NVIDIA A40 GPU in the NTU TC2 cluster. Hyperparameters are shared across runs as specified in Section III.

V. Results

All numbers in this section are from the corrected pipeline (Section IV-A) and evaluated on 299 training shapes.

TABLE II

FINAL CORRECTED POST-FIX RESULTS. VALUES ARE REPORTED EXACTLY AS USED IN THE FINAL REPORT NARRATIVE: SINGLE-SEED ROWS ARE PER-SHAPE MEANS OVER 299 TRAINING SHAPES; MULTI-SEED ROWS ARE MEANS OF PER-SEED MEANS.

ID	Config	CD (↓)	NC (↑)
Without Positional Encoding			
01	100%, no Eik	0.0361	0.729
02	100% + Eik	0.0295	0.721
03	50% + Eik	0.0288	0.748
04	10% + Eik (3-seed)	0.0315±0.0015	0.7288±0.0032
05	5% + Eik	0.0344	0.724
With Fourier Positional Encoding (10% supervision only post-fix)			
06	10% + Eik + PE $L = 6$ (3-seed)	0.0308±0.0012	0.5171±0.0066
11	10% + Eik + PE $L = 4$	0.0432	0.6090

A. Eikonal Improves Label Efficiency, but 50% Is the Clearest Operating Point

Table II summarizes the final corrected experiment set. The non-PE progression is favorable overall: the vanilla baseline (EXP-01) reaches CD = 0.0361 and NC = 0.729, while adding Eikonal at full supervision (EXP-02) improves CD to 0.0295 but slightly lowers NC to 0.721. The strongest no-PE operating point is EXP-03 at 50% supervision, which attains the best CD/NC pair among the non-PE runs (0.0288 / 0.748). Because EXP-03 was trained for 1,500 epochs while EXP-02 used 3,000 epochs, the small quantitative gap between the 50% and 100% Eikonal runs should not be interpreted as a controlled proof that 50% supervision is intrinsically better; the stronger claim is that 50% supervision remains competitive and visually reliable.

The no-PE label-efficiency trend is therefore positive but not monotone in the way one might have expected before examining the meshes. At 10% supervision, EXP-04 (3 seeds) remains numerically competitive with EXP-01 and is seed-stable (CV(CD) = 0.047), but qualitative inspection (Fig. 1) reveals mixed visual quality, with the 50% Eikonal run not uniformly better than every lower-label or non-Eikonal reconstruction on every shape. We therefore treat 50% + Eikonal as the clearest supported metric result, and 10% + Eikonal as promising but visually less reliable.

B. Fourier PE Degrades Normal Consistency at 10% Supervision

Within the 10% slice (Table II), the dominant effect of PE is on normal consistency rather than chamfer distance: EXP-06 ($L = 6$) matches the no-PE EXP-04 on CD but its NC drops from 0.729 to 0.517, reproducible across all three seeds (CV(NC) $\approx$ 0.013). EXP-11 ($L = 4$) partially recovers NC to 0.609 at the cost of CD rising to 0.0432, moving the operating point toward the no-PE baseline on NC but away from it on CD rather than producing a uniformly better PE configuration. We cannot state from corrected data that PE collapses CD across supervision ratios; we can state that, at 10% supervision, PE primarily degrades the gradient field of the learned SDF (NC) rather than the surface localization itself (CD).

C. Per-Category Breakdown

Fig. 2 breaks down the post-fix no-PE subset and the 10% PE runs into per-shape CD and NC distributions by ShapeNet category. Three observations reinforce the aggregate numbers. (i) Category difficulty is consistent in one direction: airplanes are usually the easiest on CD, while chairs and tables alternate as the hardest category depending on the run. (ii) Eikonal regularization compresses the category spread on both CD and NC (EXP-02 and EXP-03), i.e. its benefit is broadly distributed rather than concentrated in one category. (iii) The NC drop under Fourier PE (EXP-06, EXP-11) is broadly distributed across categories rather than driven by one category, consistent with PE acting on the decoder's gradient field rather than on category-specific geometry.

VI. DISCUSSION

A. Why Eikonal Enables Label Reduction

The clearest claim the corrected data support is that Eikonal regularization improves CD-based label efficiency relative to the non-Eikonal baseline. EXP-02 shows that Eikonal already improves CD at full supervision, although NC is slightly lower, and EXP-03 shows that the model can retain strong quantitative results even after halving the labels. The 10% and 5% runs remain numerically competitive, but the figures show that this should not be over-read as meaning all low-label reconstructions are equally convincing.

We attribute this to the Eikonal term acting as a volumetric geometric prior. The supervised L1 SDF loss is by construction local: it only constrains f_θ at N_s labeled points. When N_s is small (25K supervised points per shape at 10% supervision, 12.5K at 5%), the decoder has enormous freedom in unsampled regions. The Eikonal term $(\|\nabla_x f\| - 1)^2$ removes that freedom in a specific way: it requires f_θ to be an honest distance function everywhere the gradient is evaluated, including on the 250K unsupervised points per shape. Practically, this closes two failure modes that hurt low-label DeepSDF: it prevents the decoder from collapsing away from the sparse labels (as the pre-fix preprocessing caused), and it penalizes high-curvature excursions between widely-spaced labeled points. The reason the 10% + Eikonal run can match or beat 100% + no-Eikonal is not that 10% of the labels carry more signal than 100%; it is that the Eikonal term extracts geometric regularization from the other 250K unsupervised points that the non-Eikonal baseline simply discards.

Two secondary observations constrain this interpretation. First, the NC evidence is not a matched improvement story: EXP-02 has lower NC than EXP-01, and EXP-04 matches EXP-01's NC to three decimals despite using only 10% of the labels. Second, the ratio-dependent warmup schedule appears necessary in practice, although we did not run a clean ablation on warmup itself.

B. Why Fourier PE Costs Normal Consistency

The post-fix evidence at 10% supervision is that Fourier PE primarily degrades the gradient field of the learned SDF rather than surface localization. The most plausible explanation is

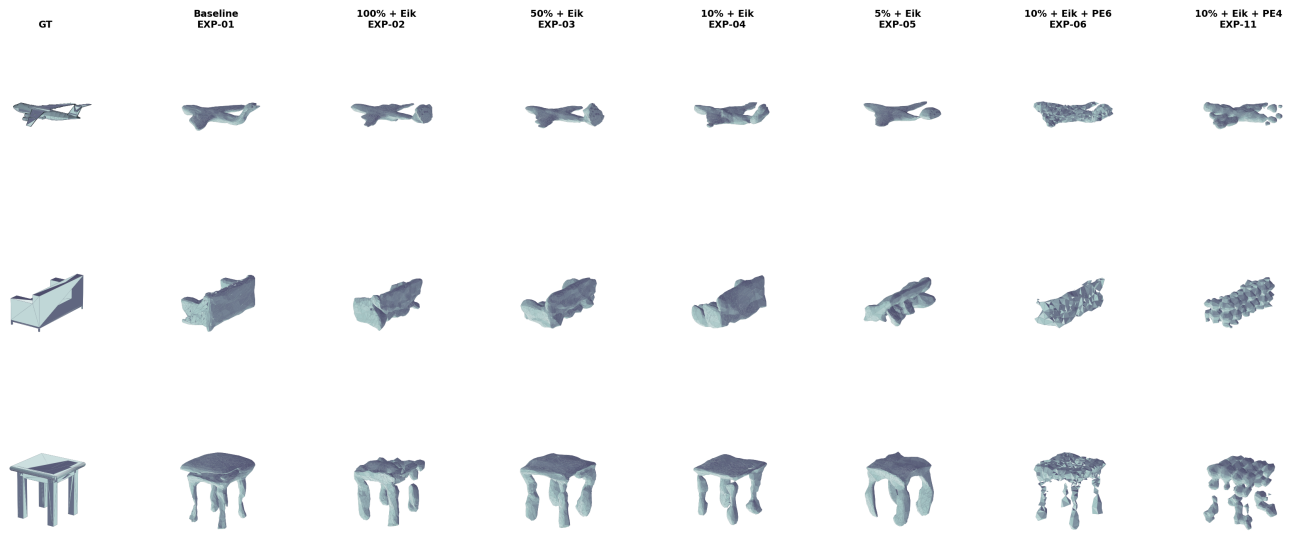

Fig. 1. Representative reconstructions by experiment. The no-PE Eikonal runs remain visually competitive with the baseline, but the figure should not be read as proving a uniform qualitative ordering across all shapes. The PE columns show the clearest qualitative degradation: thin structure disappears, surfaces become unstable, and reducing the PE frequency does not repair the failure mode.

that PE interacts badly with the training/evaluation geometry of this setup: marching cubes is clipped to the unit sphere, so the remaining failure is not cube-corner extrapolation. Instead, PE appears to induce oscillatory gradients within the evaluated unit-sphere region: the zero level set can remain close enough for CD to survive, while the surface-normal direction becomes unreliable enough for NC to degrade. Lowering the frequency from $L = 6$ to $L = 4$ partially mitigates the NC drop at the cost of higher CD; we cannot generalize this trade-off to 5% or 100% supervision from the post-fix data.

C. Limitations

Several limitations of the post-fix study should be stated plainly.

Training set evaluation only. All CD/NC numbers are computed on the 299-shape training set rather than a held-out test split. The post-fix retrains used `train_split` = 1.0 because the study is scoped around the interaction between Eikonal regularization, Fourier PE, and supervision ratio on the training objective; generalization to unseen shapes is a separate question that we do not address.

Uneven training budgets across phases. EXP-01/02 used 3,000 epochs, while the post-fix priority retrains (EXP-03/04/05/06/11) used 1,500 epochs to fit the TC2 QoS budget. This does not invalidate the observed trends, but it does make the matrix less uniform than an ideal final benchmark suite.

Single dataset and small dataset scale. We train on 299 shapes across three ShapeNet categories. Both the label-efficiency and PE-trade-off claims may behave differently at larger scale, with more categories, or on non-ShapeNet geometry.

Incomplete post-fix PE matrix. Only EXP-06 and EXP-11 (both at 10% supervision) were retrained on the corrected pipeline within the available QoS budget. The PE points at 100% and 5% supervision (EXP-09/EXP-07/EXP-10/EXP-12) and the second-order regularizer ablation (EXP-08) are not in the post-fix evaluation, so the PE conclusion is restricted to the 10% supervision slice.

Metric coverage. The original plan included volumetric IoU and an Eikonal-deviation metric ($|\|\nabla f\| - 1|$); neither was computed in the post-fix pipeline. Our conclusions are based on CD and NC only.

VII. Conclusion

We set out to test whether Eikonal regularization can reduce the amount of signed-distance supervision required by DeepSDF, and whether Fourier positional encoding helps or hurts in that low-label regime. Using only the corrected post-fix experiments, three conclusions are now clear. First, Eikonal regularization is useful: it improves the full-label baseline on CD and enables strong performance at reduced supervision. Second, the most convincing label-efficiency result is 50% supervision with Eikonal, which gives the best no-PE metrics; 10% supervision is numerically competitive but qualitatively less reliable. Third, Fourier PE is not recommended at the 10% slice we could test, with both $L = 6$ and $L = 4$ degrading NC relative to the no-PE Eikonal baseline.

The most natural next steps are therefore (i) closing the PE matrix at 100% and 5% supervision and (ii) exploring alternative high-frequency representations such as multi-resolution hash encodings [12] or revised sampling strategies that reduce the observed mismatch between training support and reconstruction queries.

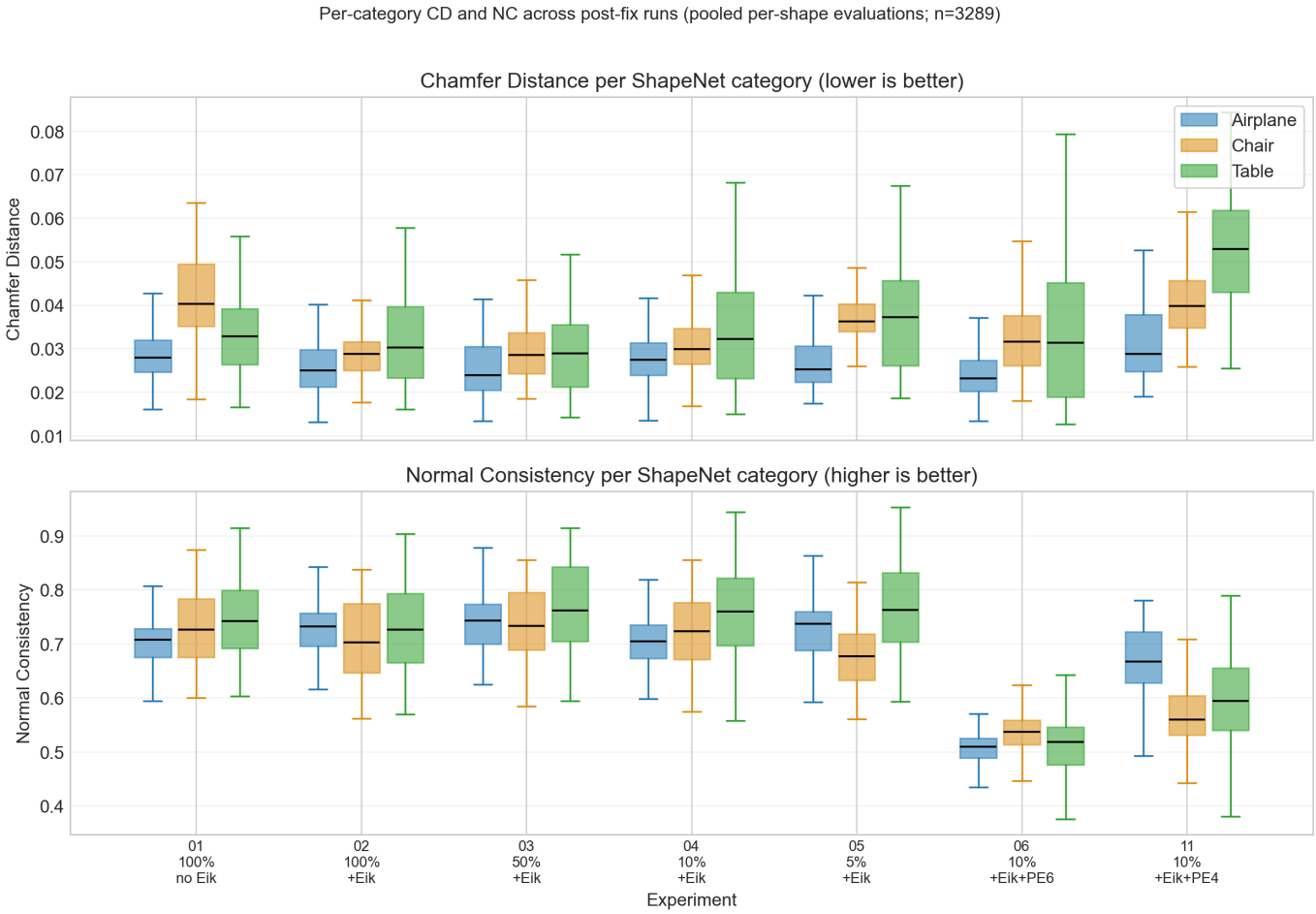

Fig. 2. Per-category subset visualization retained for qualitative support. The final report narrative uses Table II as the authoritative source for the full experiment matrix.

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